The Paradox of Opportunity

Socioeconomic Origins of Financial Gender Gaps

Athena Tsouderou and Şelale Tüzel

Discussion by Yukai Wang
Rotman Pre-seminar

October 16, 2026

Widening Gender Gaps

Common view: opportunity is an equalizer

- More income, **education**, and access should reduce gender gaps.

This paper: the opposite

- In the stock market, women's ownership gap is 2 pp (6pp) at low (high) SES.
- In the mortgage market, women pay about 0.6% (1%) more on the loan than comparable men in low (high) SES tracts.
- This creates a **paradox of opportunity**.

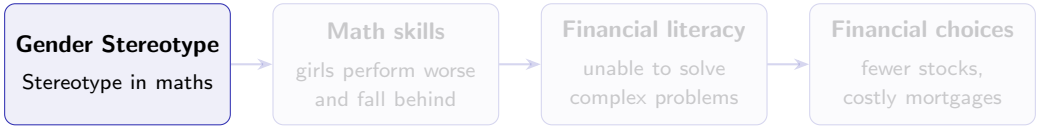
Question: How, and why, do gender gaps in financial decisions vary with socioeconomic status?

Literature

The Mechanism

Question: How, and why, do gender gaps in financial decisions vary with socioeconomic status?

"Gender differences in math are larger in advantaged environments"
(Breda et al., 2020; Dossi et al., 2021).



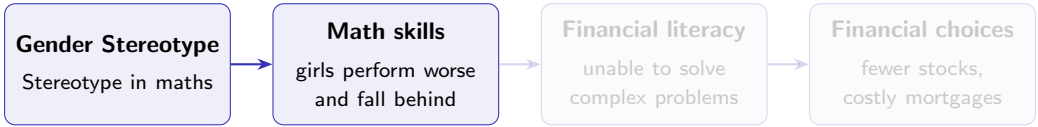
Financial choices:

- Stock market participation
- Mortgage borrowing

The Mechanism

Question: How, and why, do gender gaps in financial decisions vary with socioeconomic status?

"Gender differences in math are larger in advantaged environments"
(Breda et al., 2020; Dossi et al., 2021).



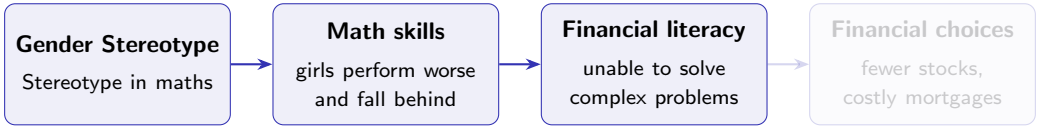
Financial choices:

- Stock market participation
- Mortgage borrowing

The Mechanism

Question: How, and why, do gender gaps in financial decisions vary with socioeconomic status?

"Gender differences in math are larger in advantaged environments"
(Breda et al., 2020; Dossi et al., 2021).



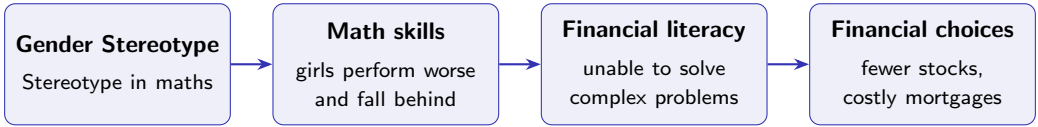
Financial choices:

- Stock market participation
- Mortgage borrowing

The Mechanism

Question: How, and why, do gender gaps in financial decisions vary with socioeconomic status?

"Gender differences in math are larger in advantaged environments"
(Breda et al., 2020; Dossi et al., 2021).



Financial choices:

- Stock market participation
- Mortgage borrowing

Stock Ownership

1. **PSID Family Files**
   - Single adult household heads data
   - Financial literacy dummy from one question
   - Income, education, stock ownership, ...

2. **CDS Child-to-Adult Panel**
   - Children who were ages 0–12 in 1997 and across three waves
   - Covers **ability** and **attitude** in math and reading
   - compares brothers and sisters raised in the same family
   - The children become observations in PSID once they reach adulthood

PSID Summary Stats CDS Summary Stats

Stock Ownership: Sibling Comparisons

	Stock ownership		Math decile	Math attitude	Math decile
	(1)	(2)	(3)	(4)	(5)
Female	−0.032***	−0.033***	0.108	−0.116	−0.105
	(0.011)	(0.011)	(0.195)	(0.073)	(0.106)
Female × SES	−0.102***	−0.084**	−0.939**	−0.331**	
	(0.036)	(0.037)	(0.391)	(0.143)	
Top math		0.067**			
		(0.031)			
Top reading		0.020			
		(0.032)			
Math test decile				0.111***	
				(0.016)	
Lagged math decile					0.550***
					(0.028)
Lagged math attitude					0.119***
					(0.042)
Observations	8,638	8,638	5,616	2,604	1,599

Adult Result Comment 1 Story

Mortgage: Data

1. **Merged HMDA with Fannie Mae and Freddie Mac**
   - Conventional loans to single borrowers, originated 2018-2019
   - Each loan tracked monthly until September 2023 for refinancing and default
   - Rate, fees, credit score, ...
2. **ACS at Census tract level**
   - Identify SES of each mortgage under tracts level
   - Offer education, income, etc...
3. **Local math gaps, matched to each loan's location**
   - Math score gap: in math scores, grades 3-8, by school district (SEDA)
   - Job math gap: in job math intensity, by tract (O*NET and ACS)

Mortgage: Result with Maths

	Overall cost (% of loan, first 10 years)								
	No controls			Pricing controls			+ Additional controls		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Female	0.606***	0.807***	0.757***	0.627***	0.845***	0.752***	−0.025	0.021**	0.045***
	(0.017)	(0.024)	(0.014)	(0.016)	(0.020)	(0.013)	(0.015)	(0.011)	(0.012)
Female × SES	0.445***			0.367***			0.207***		
	(0.028)			(0.026)			(0.024)		
Female × Math score gap		0.183***			0.148***			0.045***	
		(0.021)			(0.018)			(0.012)	
Female × Job math gap			0.177***			0.147***			0.086***
			(0.014)			(0.013)			(0.012)
Location × month FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Pricing controls	No	No	No	Yes	Yes	Yes	Yes	Yes	Yes
Additional controls	No	No	No	No	No	No	Yes	Yes	Yes
Observations	1,826,135	2,159,375	1,826,137	1,826,135	2,159,375	1,826,137	1,706,668	2,044,464	1,706,670

Baseline

Alternative Explanations

Stocks: the widening gap remains after accounting for:

- Wealth and risk tolerance.
- Expected future income and household size.
- Dropping the poorest quarter of households.

Mortgages: the widening gap is not explained by:

- Credit risk: women default about 20% less.
- Lenders: gaps hold within the same lender, and refinance denials are equal by gender.
- Childcare: the gender gap in caregiving is smaller in high-SES areas.

Discussion

My overall takeaway

Very good design!

- The authors worked a lot to find measurements of math skills.
- The pattern holds in two very different decisions.
- Trace children's growth until they reach adulthood.

My comments:

- Opportunity may reveal the gap rather than create it
- Does math build financial literacy, or run alongside it?
- "Who wears the pants?" (if time permits)

Comment 1: Does opportunity reveal the gap or create it?

- The authors address the concern that low-SES people face limited choice sets.
- However, acting may have a **fixed cost**.

1. **At low-SES**:
   - The stakes are too small for the gain to cover the cost
   - Both men and women are below the threshold
2. **At high-SES**:
   - stakes push men over the threshold sooner than women
   - the same difference shows as a large gap

Comment 1: Does opportunity reveal the gap or create it?

Story 1: the underlying gap itself grows with SES (the paper's view)

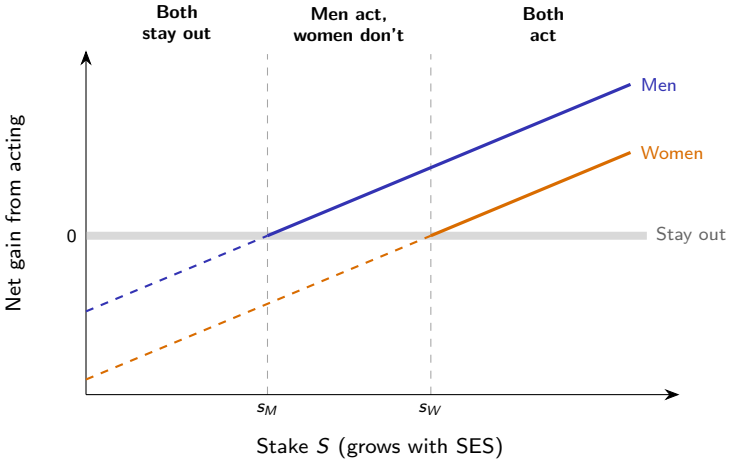
- The gap is "created" by SES
- It runs through education and environment.
- Yet childhood math explains only a fifth of the sibling gap.

Story 2: the underlying gap reveals a gender difference that exists everywhere

- The gap is fixed across SES, but vary with gender.
- Agents only act when the stakes outweigh the fixed cost.
- The gap in high SES can be explained by the higher fixed cost.

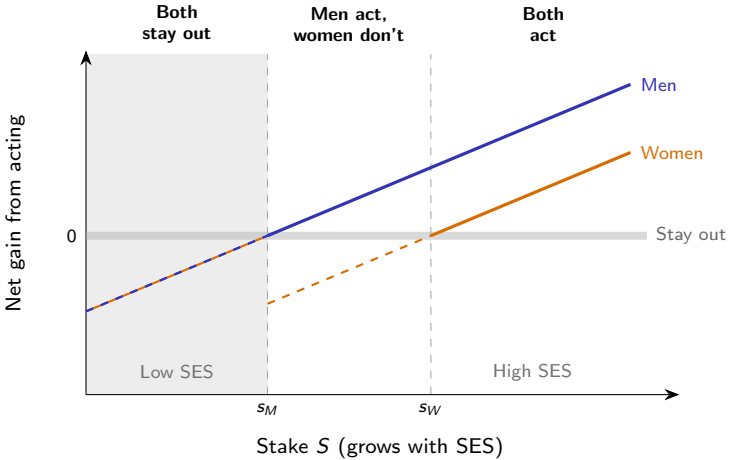
Sibling Regression

Comment 1: Does opportunity reveal the gap or create it?



Story 2: the same cost gap at every SES level

Comment 1: Does opportunity reveal the gap or create it?



Story 1: no cost gap at low SES, a gap at high SES

Comment 1 (cont): How to test both stories

For the authors:

- Let the gender gap vary with wealth, not only with SES.
- Interact wealth with gender.
- Decompose into two gradients: wealth (Story 2) and education (Story 1).

Takeaway:

- Both stories can be true; the question is how much of each.
- A small gap at low SES could mean nobody moves.

Comment 2: Math literacy or financial literacy?

Definitions of financial literacy are broad:

- Lusardi & Mitchell (2014): the **ability** to process economic information and **make informed decisions** about saving, debt and retirement.
- This paper: the ability to recognize, evaluate, and act on financially valuable opportunities.

Measurements are also broad:

- Lusardi & Mitchell (2014): multiple choice questions of compounding, inflation and diversification
- This paper uses one open-ended calculation in PSID.
- Question: which parts of the definition do these measures capture?

LM Question

Comment 2 (cont): The measurement

This paper uses one question:

- "Suppose you have $200 in a savings account. The account earns 10 percent interest each year. How much would you have in the account at the end of two years?"
- Yet there are many ways to make mistakes.

Interpretations of wrong:

- $240: doesn't know compounding (**knowledge**)
- $244: arithmetic problem (**math**)
- $42: computed the interest, not the balance (**reading**)
- Blank: unsure, or not trying (**confidence** or effort)

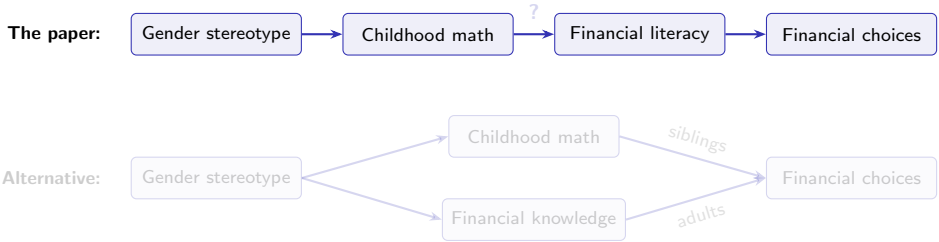
All these errors are recorded zero...

Other Questions

Comment 2 (cont): Sequential or parallel?

Two claims about the compound-interest question:

- **Narrow**: the question captures the numeracy part of literacy.
- **Broad**: the question correlates with broader indices, capturing overall financial knowledge.

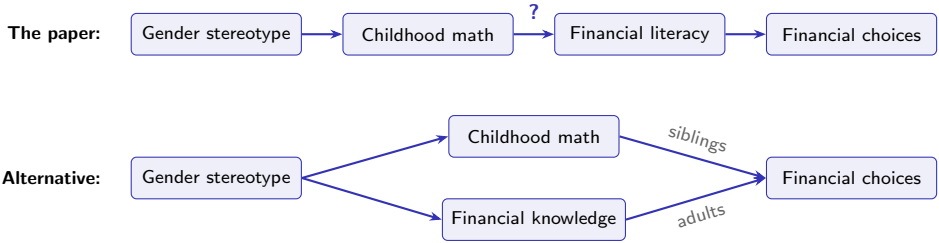


The evidence fits both!

Comment 2 (cont): Sequential or parallel?

Two claims about the compound-interest question:

- **Narrow**: the question captures the numeracy part of literacy.
- **Broad**: the question correlates with broader indices, capturing overall financial knowledge.



The evidence fits both!

Comment 2 (cont): Reconcile the two claims

If possible, put pure math and literacy in the same regression:

- Use the survey's quantitative-reasoning section as pure math ability.
- If math absorbs literacy's effect: points to **sequential**, as the paper argues.
- If literacy keeps its own effect: points to **parallel**, financial knowledge beyond math.

This could tell different implications:

- Sequential means more math class
- Parallel means more "personal finance" class

Extra: Who wears the pants?

– Would it be helpful to separate intrahousehold problem and individual problem?

Who Wears the Pants? Gender Identity Norms and Intrahousehold Financial Decision-Making

DA KE*

ABSTRACT

Using microdata from U.S. household surveys, I document that families with a financially sophisticated husband are more likely to participate in the stock market than those with a wife of equal financial sophistication. This pattern is best explained by gender identity norms, which constrain women's influence over intrahousehold financial decision-making. A randomized controlled experiment reveals that female identity hinders idea contribution by the wife. These findings underscore the roles of intrahousehold bargaining and traditional norms in shaping household financial decisions.

Conclusion

Conclusion

- This paper documents the paradox of opportunity: the gender gap is larger among high-SES groups.
- The mechanism is the stronger "math is for boys" in advantaged families.
- The math skills bias shapes their future financial literacy.
- My suggestion:
  - think about choice theory in different angles
  - be more robust in skills vs. financial knowledge, confidence, etc.

Thank you!

Stock Ownership: Single Adult

	Stock ownership						
	(1)	(4)	(6)	(7)	(8)	(9)	(10)
Female	−0.034***	−0.021***	−0.017***	−0.026***	−0.024***	−0.020**	−0.011
	(0.005)	(0.004)	(0.003)	(0.006)	(0.006)	(0.009)	(0.009)
Female × SES		−0.041***	−0.043***	−0.035**	−0.036**	−0.068***	−0.048**
		(0.011)	(0.010)	(0.015)	(0.015)	(0.023)	(0.023)
Safe assets			0.038***				
			(0.007)				
Risk tolerance					0.079***		
					(0.022)		
Financial literacy							0.130***
							(0.017)
Sample	All	All	Safe assets	Risk tolerance	Risk tolerance	Fin. literacy	Fin. literacy
Observations	56,395	56,395	55,246	29,358	29,358	8,786	8,786

Back

Mortgage: Baseline Result

	Mortgage rate (%)				
	(1)	(2)	(5)	(8)	(9)
Female	0.028***	0.019***	0.034***	0.005***	0.004***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Female × SES		0.026***	0.020***	0.010***	0.009***
		(0.002)	(0.001)	(0.001)	(0.001)
	Net loan cost (% of loan)				
	(10)	(11)	(14)	(17)	(18)
Female	0.166***	0.151***	0.109***	−0.009***	−0.007***
	(0.002)	(0.003)	(0.003)	(0.002)	(0.002)
Female × SES		0.043***	0.036***	0.012***	0.014***
		(0.004)	(0.004)	(0.004)	(0.003)
Pricing controls	No	No	Yes	Yes	Yes
Additional controls	No	No	No	Yes	Yes
Lender FE	No	No	No	No	Yes

Back

Summary Stats: PSID Adult

	Low SES		High SES	
	Female	Male	Female	Male
Number of unique adults	6,277	5,419	3,260	3,093
Number of observations	27,379	18,431	11,366	9,045
Age (years)	46.66	39.84	44.19	40.01
Income ($000)	22.31	31.77	63.13	77.72
Education (years)	11.90	11.79	15.19	15.14
Stock ownership (dummy)	0.05	0.05	0.17	0.22
Safe assets	50,520	53,834	139,175	198,594
Risk tolerance	0.27	0.29	0.28	0.30
Financial literacy	0.11	0.15	0.28	0.51

Back

Summary Stats: CDS

	Low SES		High SES	
	Female	Male	Female	Male
Number of unique children	704	750	235	234
Age in childhood (years)	11.44	11.27	11.43	11.32
Parental income ($000)	40.95	42.44	106.89	105.90
Parental education (average years)	12.19	12.28	15.63	15.55
Math test decile	4.83	4.99	7.05	7.57
Reading test decile	5.33	4.57	7.16	6.69
Math attitude (standardized)	−0.07	0.00	−0.10	0.31
Reading attitude (standardized)	0.16	−0.19	0.23	−0.13
Number of observations (adulthood)	3,287	3,168	1,150	1,064
Age in adulthood (years)	23.87	23.99	23.79	23.75
Stock ownership (dummy)	0.04	0.07	0.17	0.23

Back

Contribution

- Women hold fewer stocks than men, on average.
  Jianakoplos & Bernasek (1998); Barber & Odean (2001); Bacher (2025)
  ⇒ **This paper:** the gap is near zero at low SES and grows with SES.

- Quantitative skills, more than verbal ones, drive financial decisions.
  Agarwal & Mazumder (2013); Lusardi & Mitchell (2014)
  ⇒ **This paper:** math skills formed in childhood help explain why gaps widen with SES.

- Many borrowers overpay for mortgages and fail to refinance.
  Agarwal, Rosen & Yao (2016); Keys, Pope & Pope (2016); Bhutta, Fuster & Hizmo (2026)
  ⇒ **This paper:** women pay more and refinance less, mostly in high-SES areas.

- Math gender gaps are larger in advantaged settings.
  Reardon et al. (2019); Breda et al. (2020); Dossi et al. (2021)
  ⇒ **This paper:** these gaps carry over into household finance.

Lusardi & Mitchell (2014) Financial Literacy

1. Suppose you had $100 in a savings account and the interest rate was 2% per year. After 5 years, how much do you think you would have in the account if you left the money to grow? **[Compound]**
2. Imagine that the interest rate on your savings account was 1% per year and inflation was 2% per year. After 1 year, how much would you be able to buy with the money in this account? **[real interest]**
3. Please tell me whether this statement is true or false. "Buying a single company's stock usually provides a safer return than a stock mutual fund." **[Diversification]**

The choices are usually: somewhat above, somewhat below, correct, or I don't know.

Back

Other Questions

1. You buy a drink for 85 cents and pay with $1. How much change should you get? **[15 cents]**
2. A $300 sofa is on sale at half price. How much does it cost? **[$150]**
3. The chance of getting a disease is 10%. Out of 1,000 people, how many would get it? **[100]**
4. A used car sells for $6,000, which is two-thirds of its price new. What did it cost new? **[$9,000]**
5. Five people share a $2 million lottery prize. How much does each get? **[$400,000]**

Back